DIODE

THEORY OF OPERATION

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TOPIC OUTLINE

The Unbiased Diode

The Biased Diode

- Reverse Bias and Forward Bias

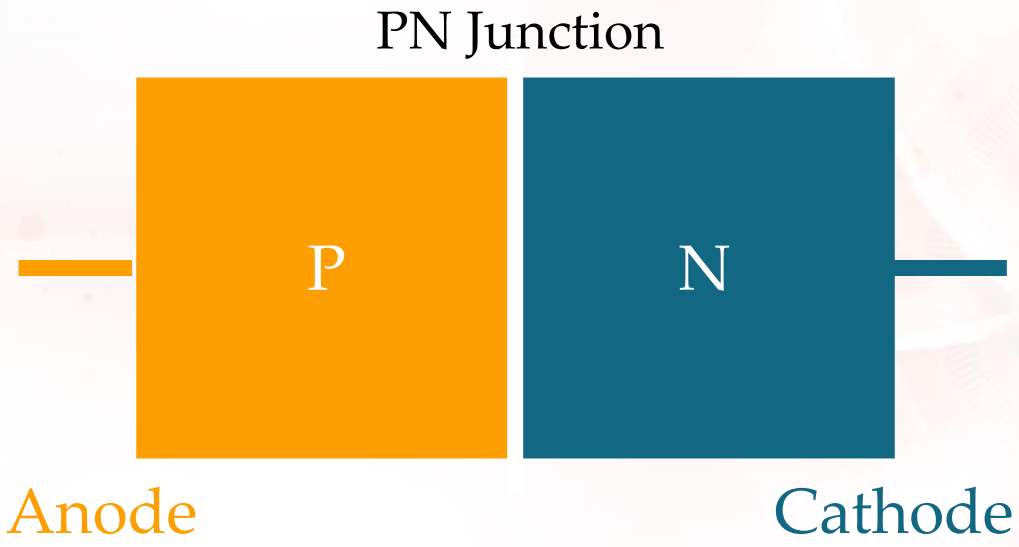
Diode Parameters

Rectifier Circuits

- Half-Wave and Full-Wave

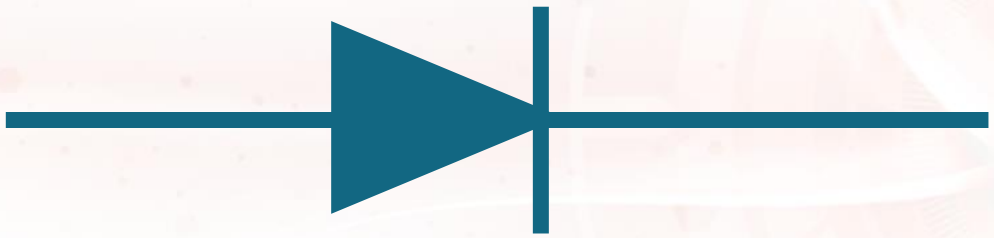
THE UNBIASED DIODE

Basic Construction



The **boarder** between the P-type and N-type material

Schematic Symbol



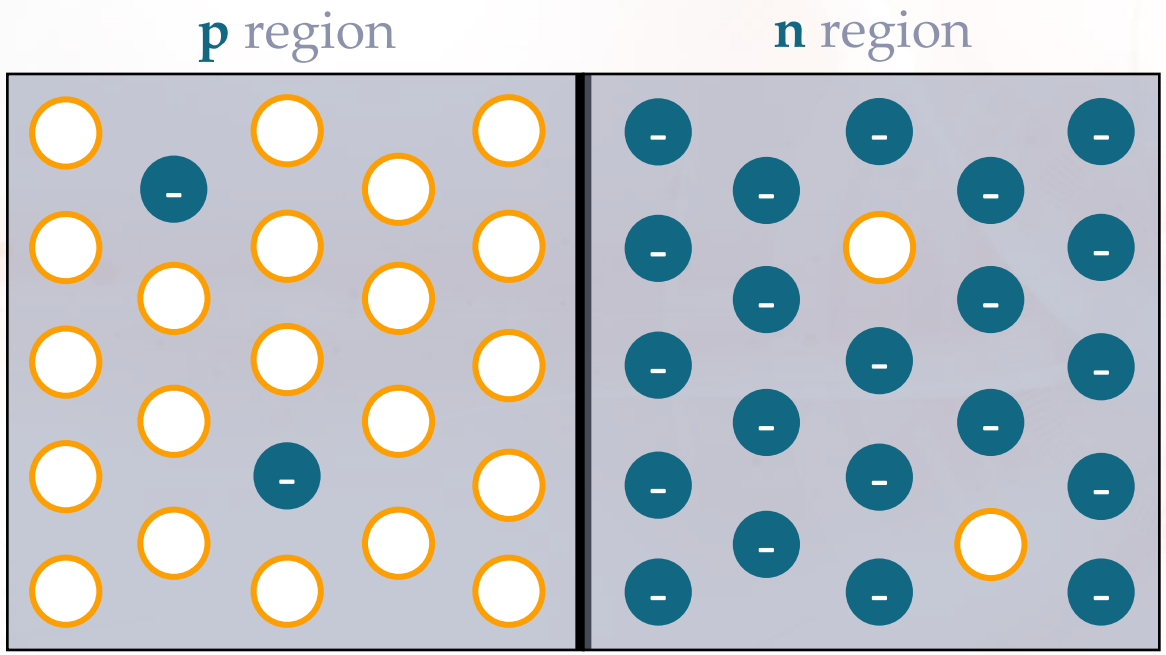
Anode

Cathode

"**Di**" stands for two

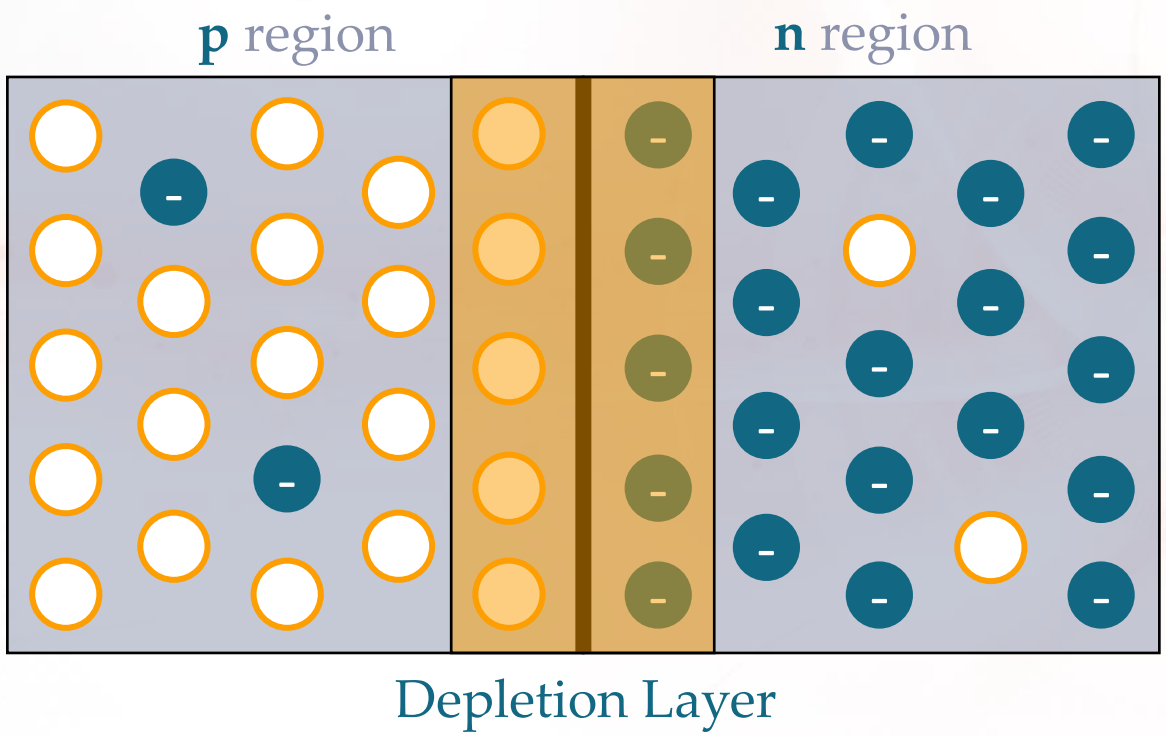
Construction of **two electrode**

DEPLETION LAYER



PN Junction

DEPLETION LAYER



Recombination

the process in which electrons and holes combine

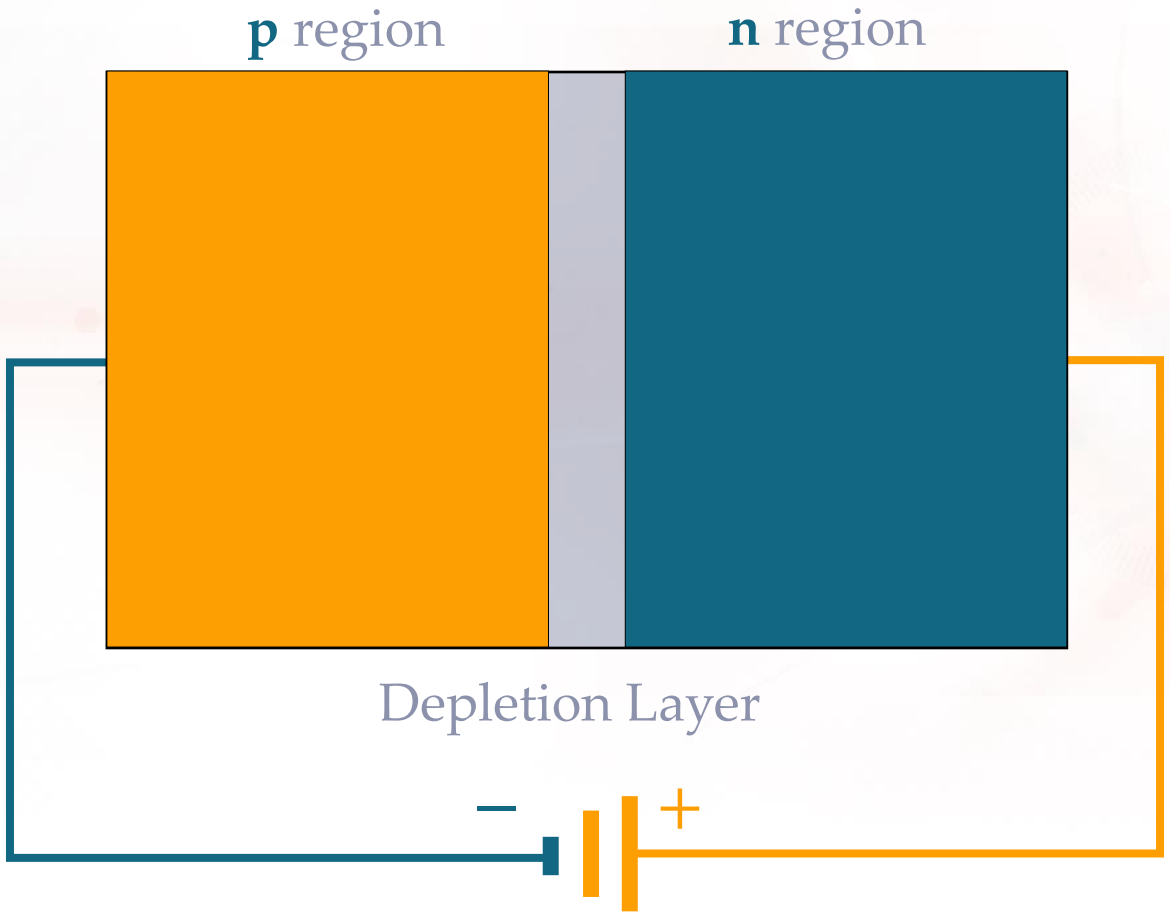
Barrier Potential

The electric field between the ions

0.3V for Germanium (Ge)

0.7V for Silicon (**Si**)

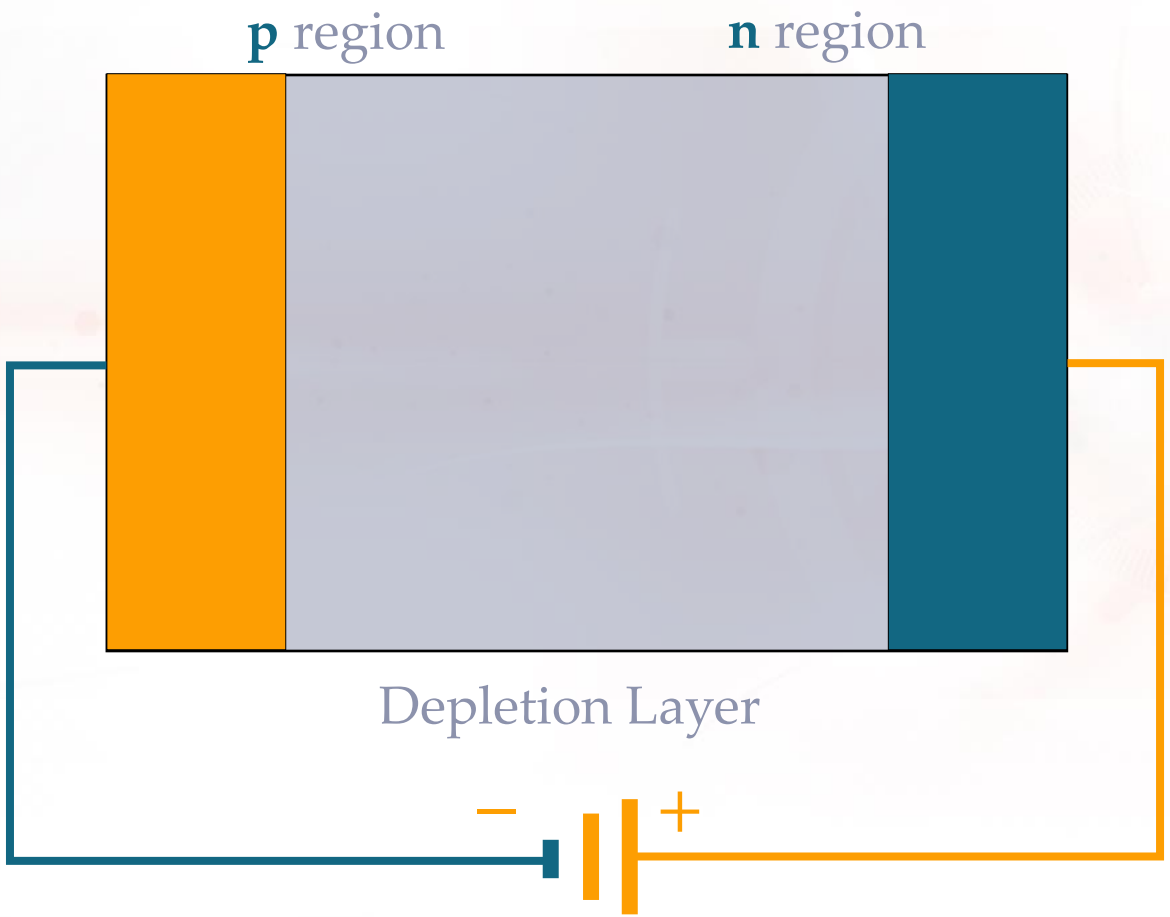
THE BIASED DIODE



Bias

application of **power source**

REVERSE BIAS



Bias

application of **power source**

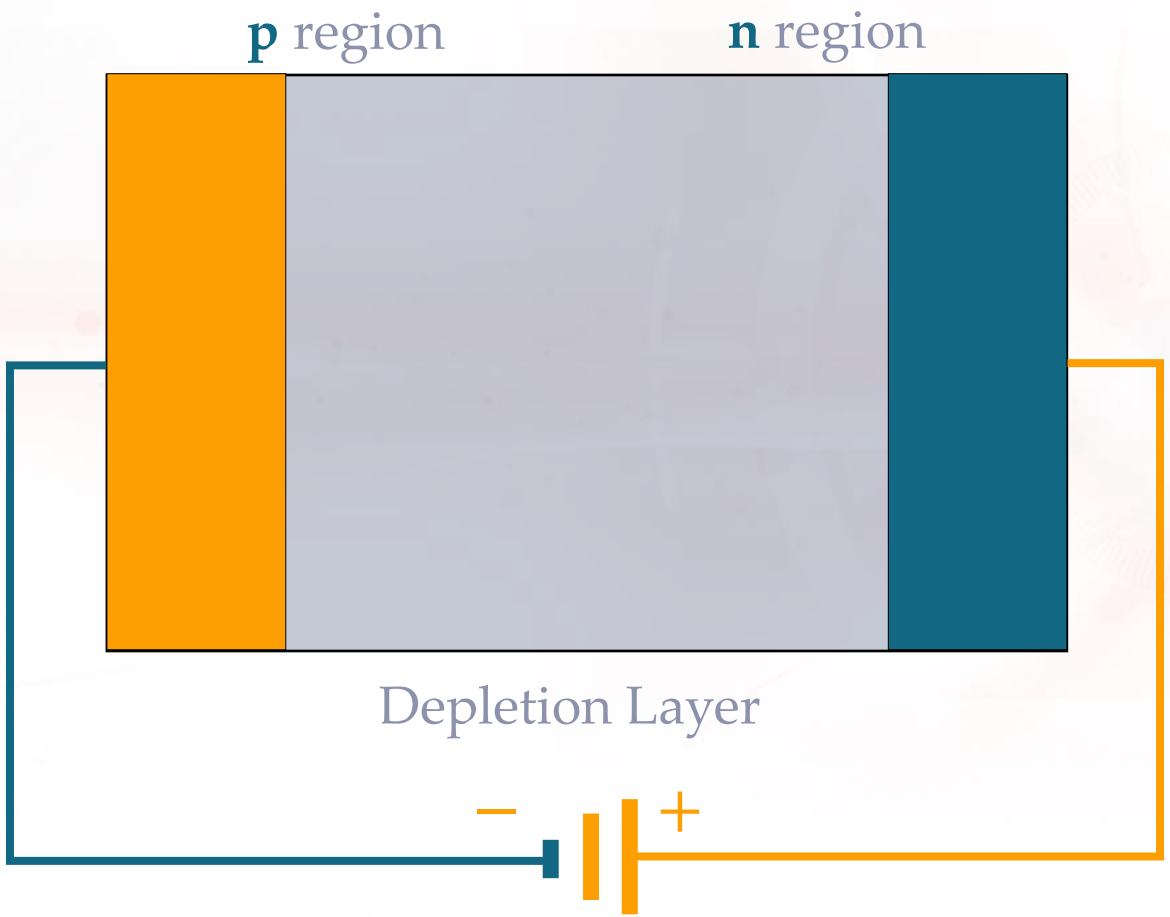
Reversed Saturation Current

The **reversed current** caused by the thermally produced minority carriers

Surface-leakage Current

Caused by the surface **impurities** and **imperfections** in the crystal structure

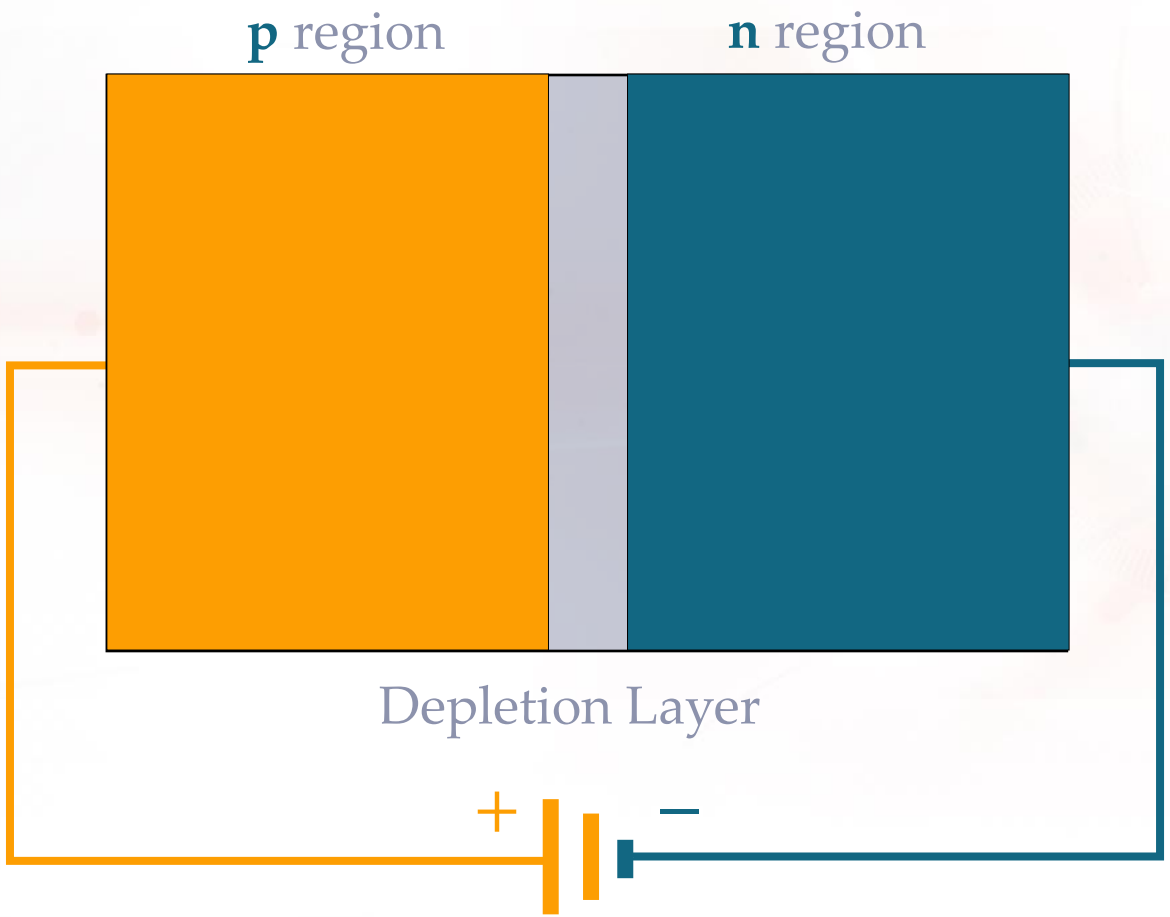
BREAKDOWN VOLTAGE



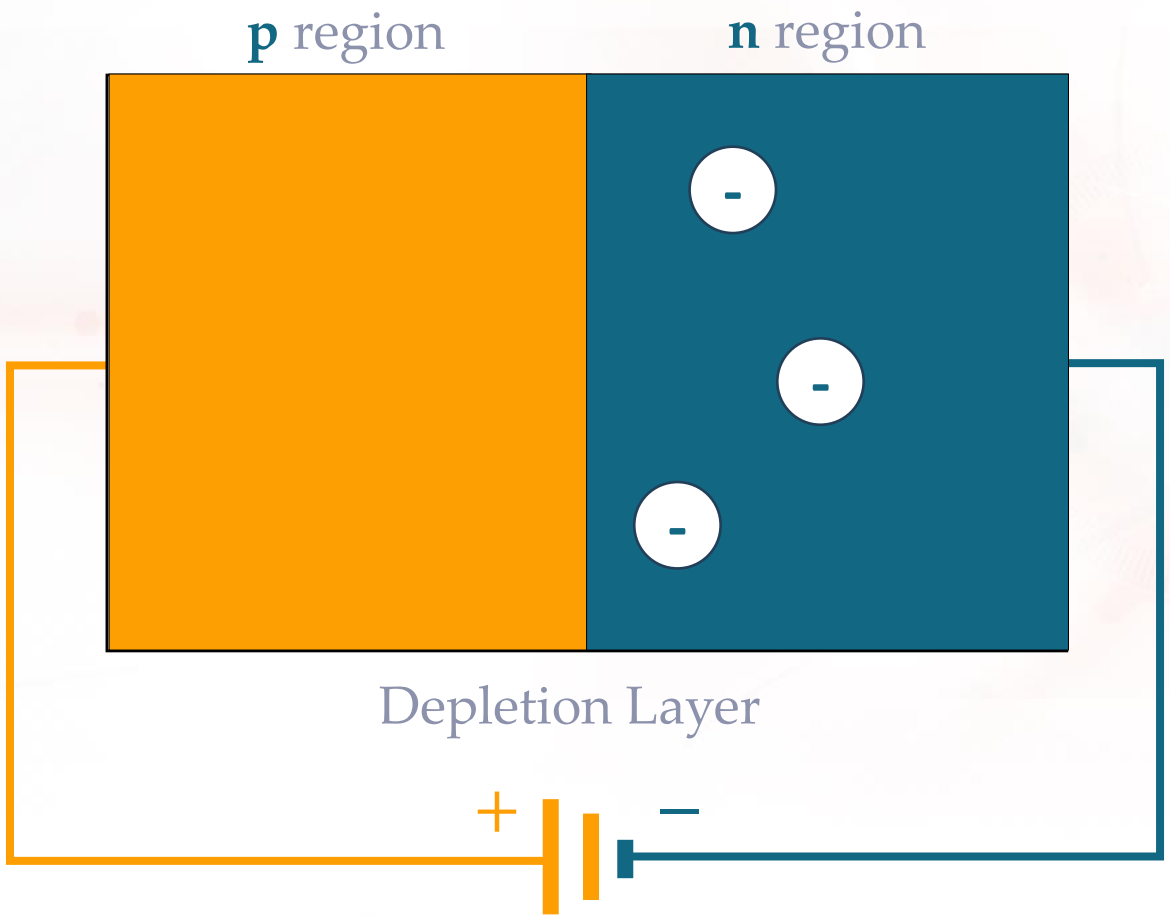
The maximum reverse voltage a diode can withstand before it is **destroyed**

50V (typ)

FORWARD BIAS



FORWARD BIAS

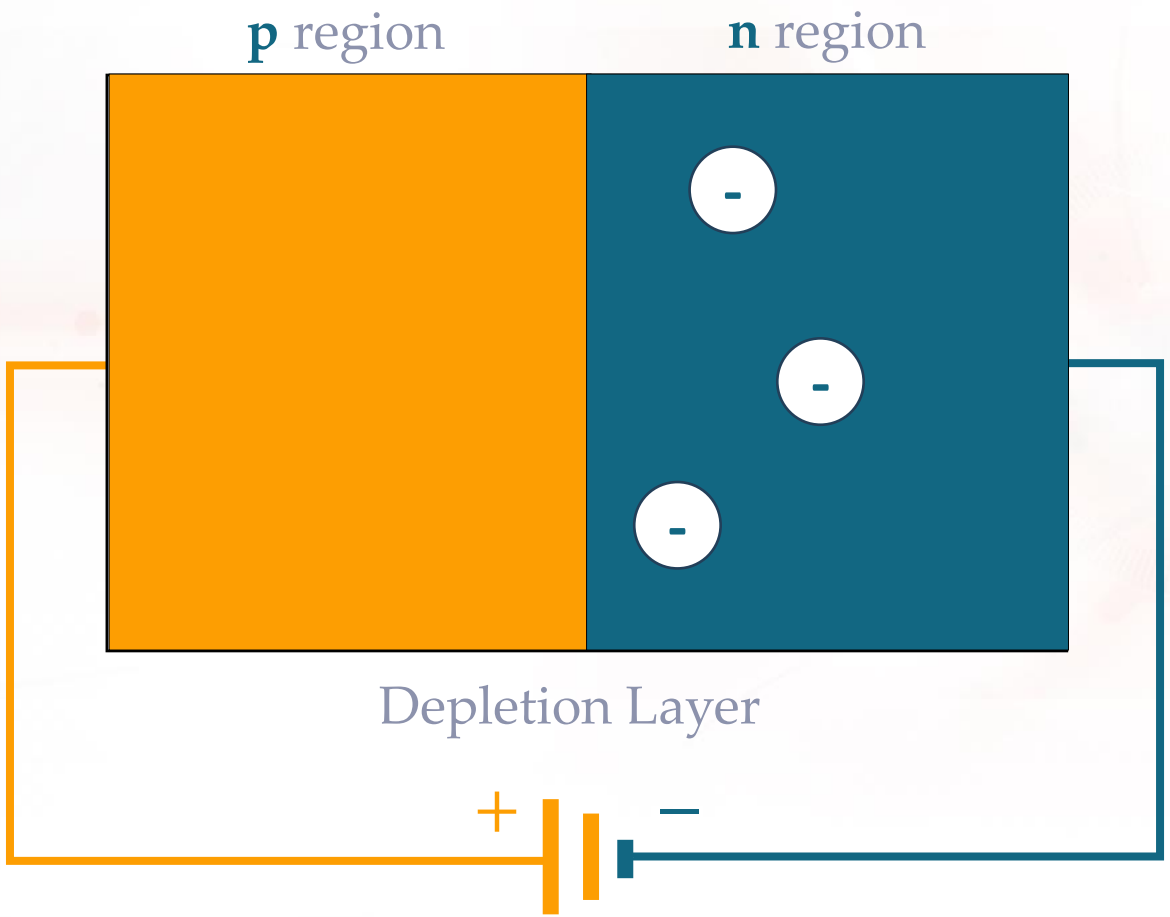


Current flows easily as long as the applied voltage is **greater** than the barrier potential

0.3V (Ge)

0.7V (Si)

FORWARD BIAS



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0.7V (Si)

DIODE PARAMETERS

TEMPERATURE

Ambient Temperature

Air **outside** the diode

Junction Temperature

Inside the diode

@ Conducting

Junction Temp > Ambient Temp

BARRIER POTENTIAL

Depends on the junction temperature

$$\frac{\Delta V}{\Delta T} = -2\text{mV}/^{\circ}\text{C}$$

EXERCISE

Assuming a barrier potential of **0.7 V** at an ambient temperature of **25°C**. Determine the **barrier potential** of a silicon diode when the junction temperature is

- **100°C**
- **0°C**
- **50°C**

REVERSE SATURATION CURRENT, I_s

I_s **doubles** for each 10°C rise

$$\%\Delta I_s = 100\% \text{ per } 10°C$$

The higher the junction temperature, the greater the saturation current

If the change in temperature are less than 10°C

$$\%\Delta I_s = 7\% \text{ per } °C$$

EXERCISE

A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

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@100°C,

25°C	**5** nA
35°C	**10** nA
45°C	**20** nA
55°C	**40** nA
65°C	**80** nA
75°C	**160** nA
85°C	**320** nA
95°C	**640** nA
96°C	$(640\ nA) + (640\ nA)(0.07)$

doubles

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doubles

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@100°C,

25°C	5 nA	
35°C	10 nA	
45°C	20 nA	
55°C	40 nA	doubles
65°C	80 nA	
75°C	160 nA	
85°C	320 nA	
95°C	640 nA	
96°C	684.8 nA	
97°C	732.7 nA	
98°C	784.0 nA	7%
99°C	838.9 nA	
100°C	$\boxed{897.6\ nA}$	

ans

EXERCISE

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@100°C,

EXERCISE

A silicon has a saturation current of **5nA** at **25°C**. What is the **saturation current** at **100°C**? At **80°C**?

@80°C,

SURFACE-LEAKAGE CURRENT, I_{SL}

Directly proportional to the **reverse voltage**

Surface-Leakage Resistance, R_{SL}

$$R_{SL} = \frac{V_R}{I_{SL}}$$

EXERCISE

If the surface leakage current is **2nA** for a reverse voltage of **25V**, what is the **surface-leakage current** for a reverse voltage of **35V**? of **100V**?

RECTIFIER CIRCUITS

HALF-WAVE RECTIFIER

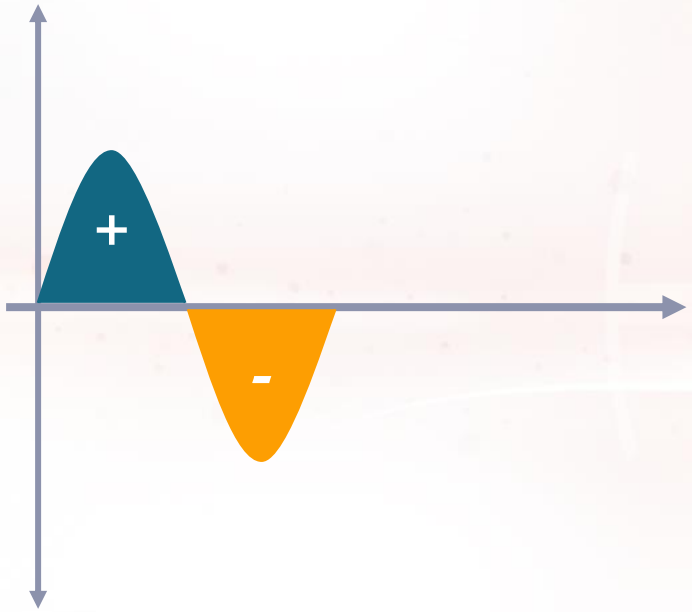
Input

Forward-Biased

Output

HALF-WAVE RECTIFIER

Input

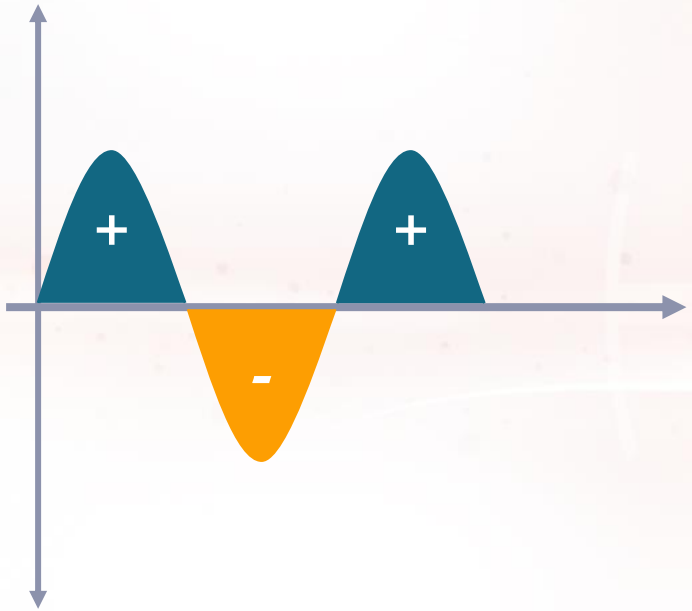


Reverse-Biased

Output

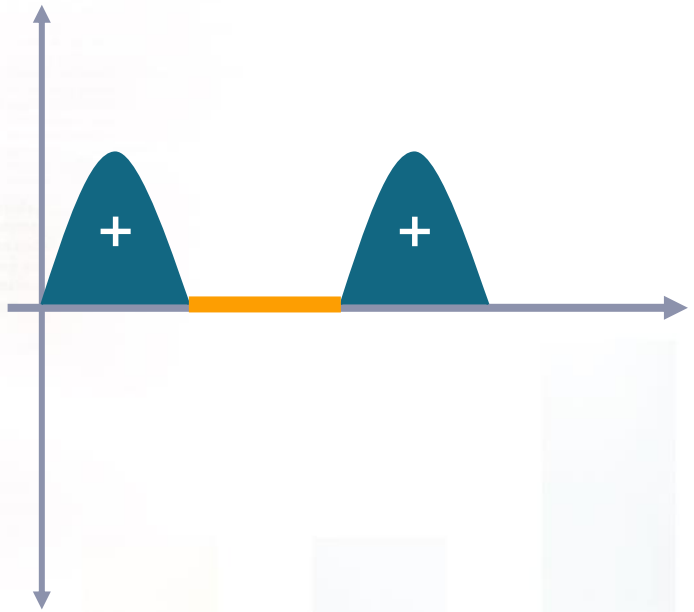
HALF-WAVE RECTIFIER

Input



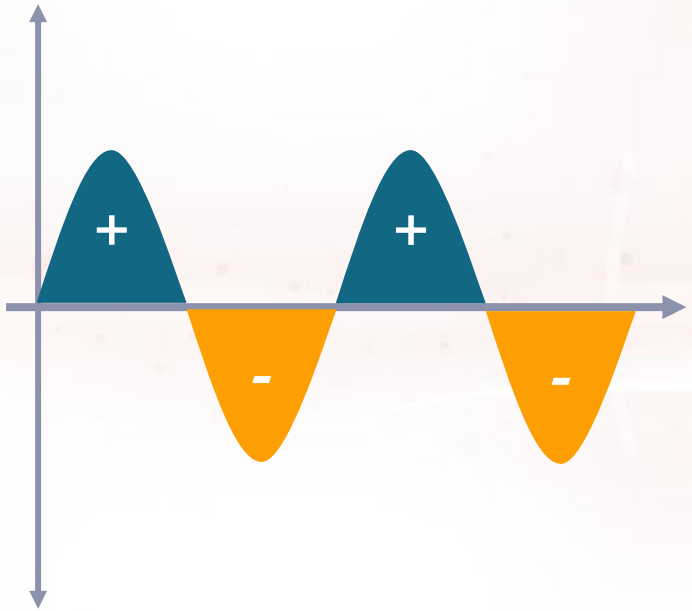
Forward-Biased

Output



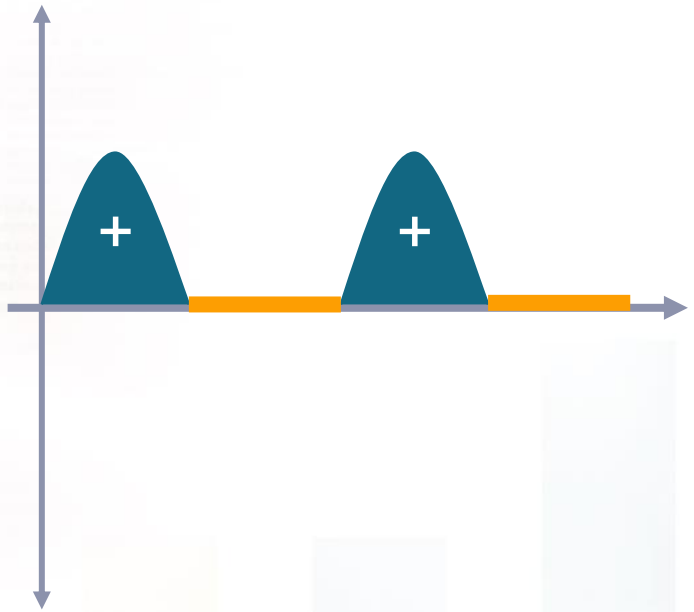
HALF-WAVE RECTIFIER

Input



Reverse-Biased

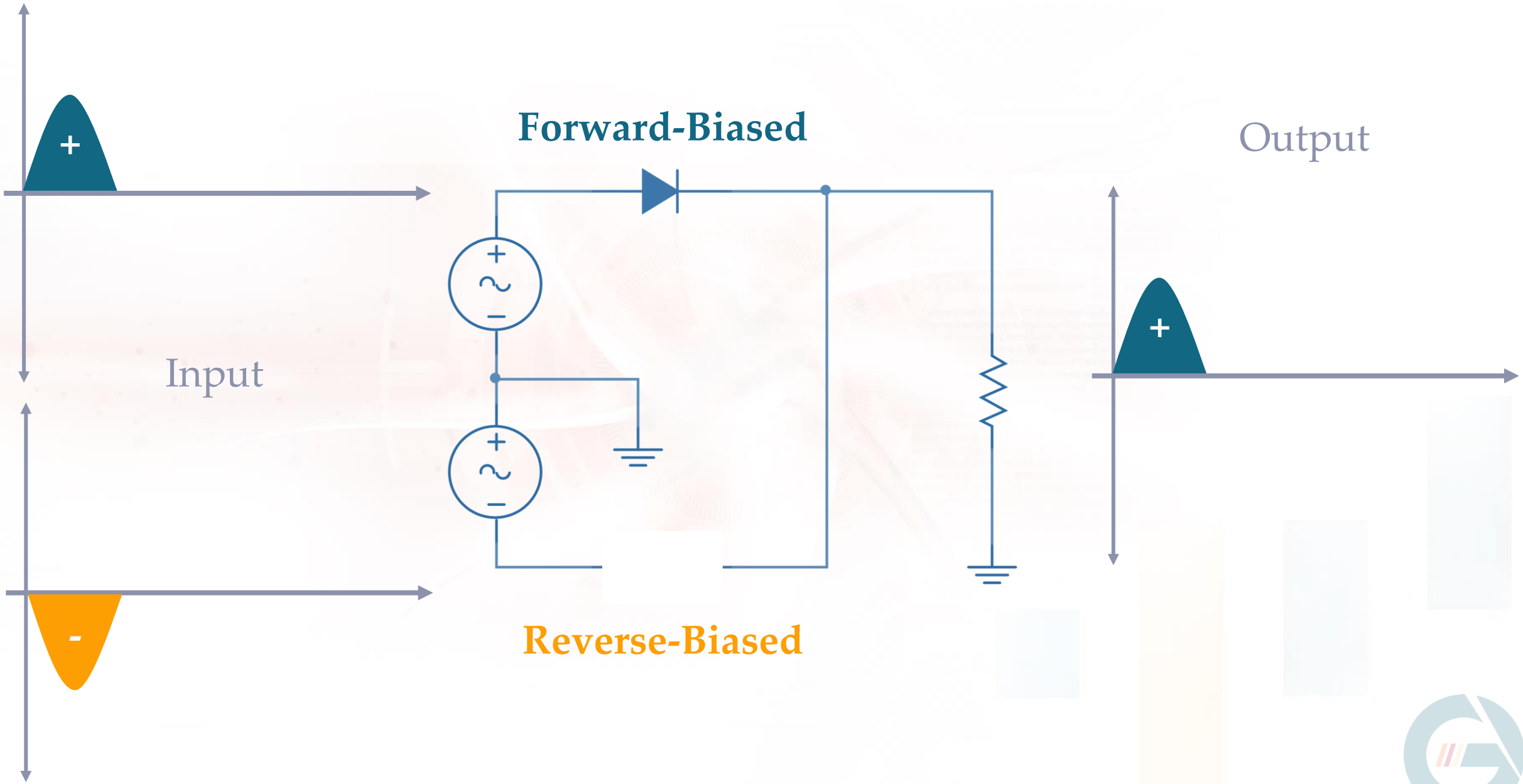
Output



Rectifier

A circuit that converts **AC** voltage to **DC** voltage

FULL-WAVE CENTER-TAPPED RECTIFIER



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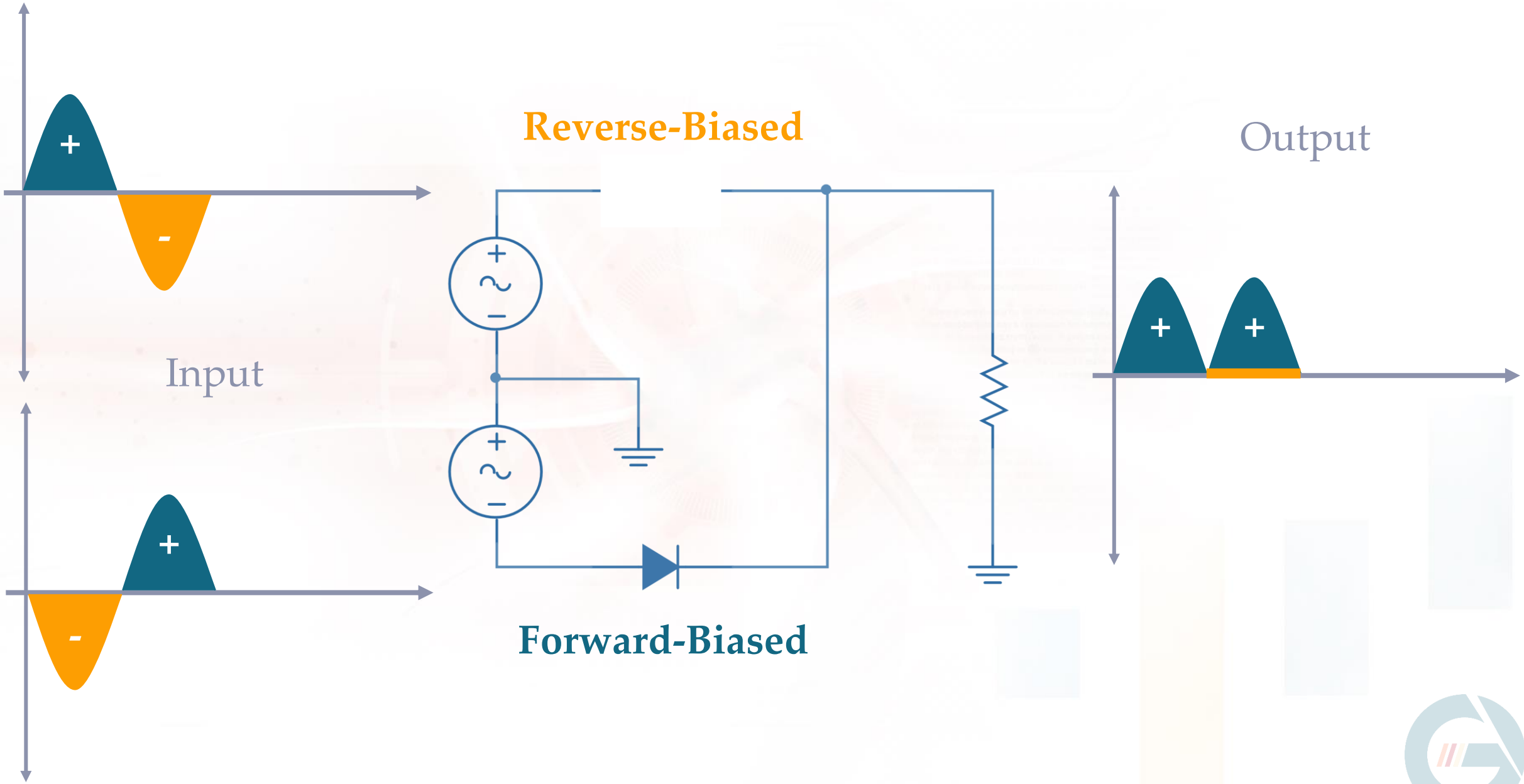

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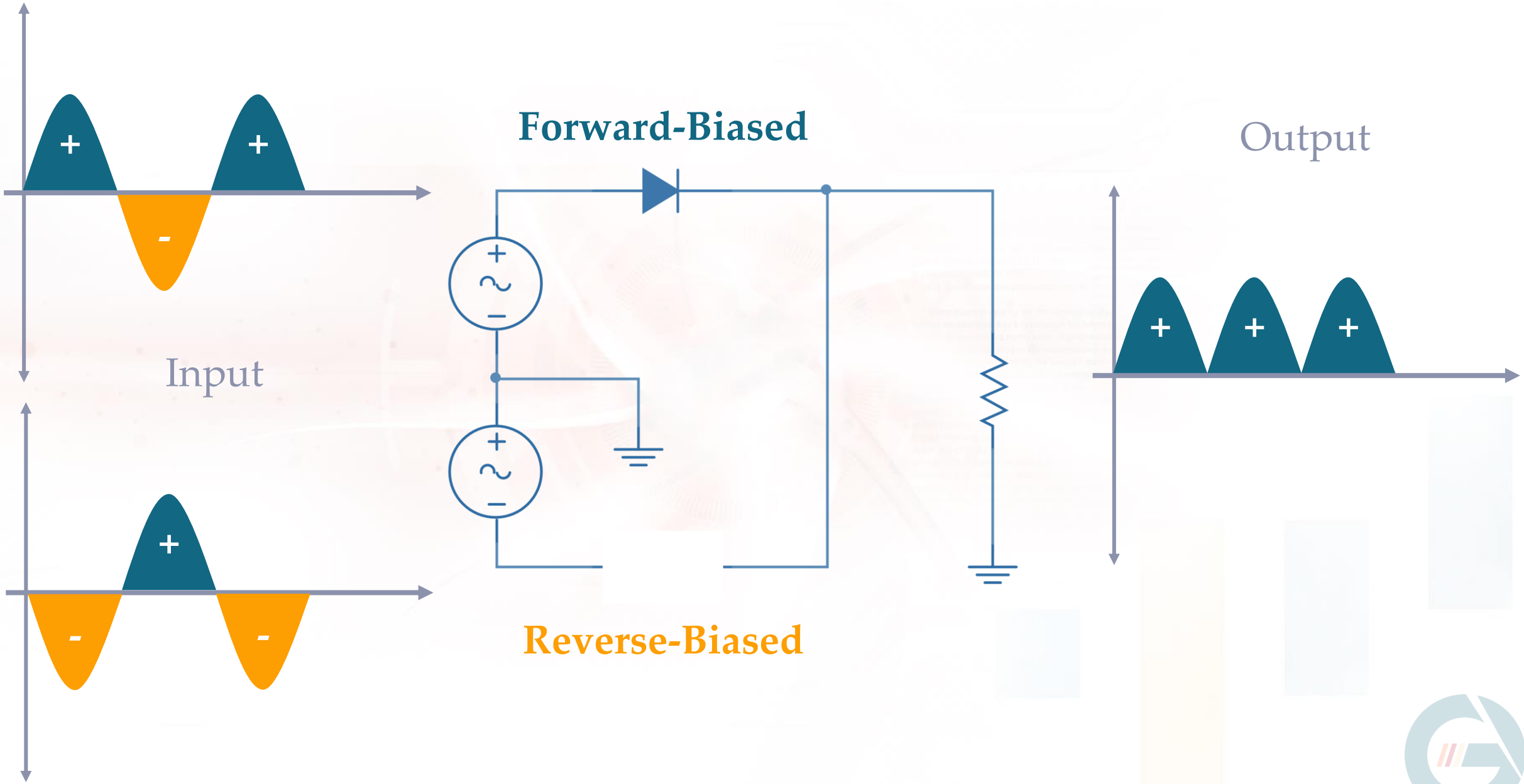

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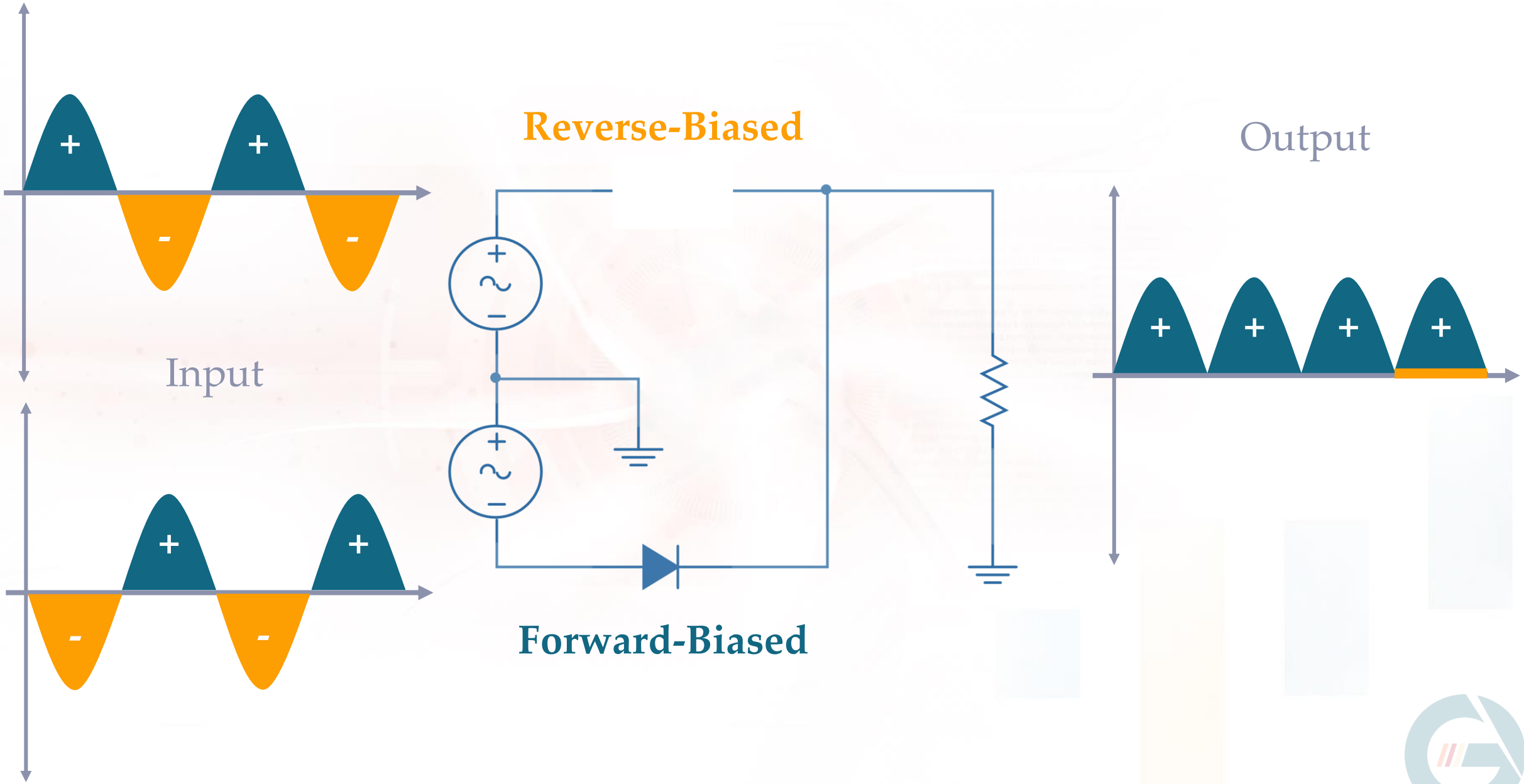
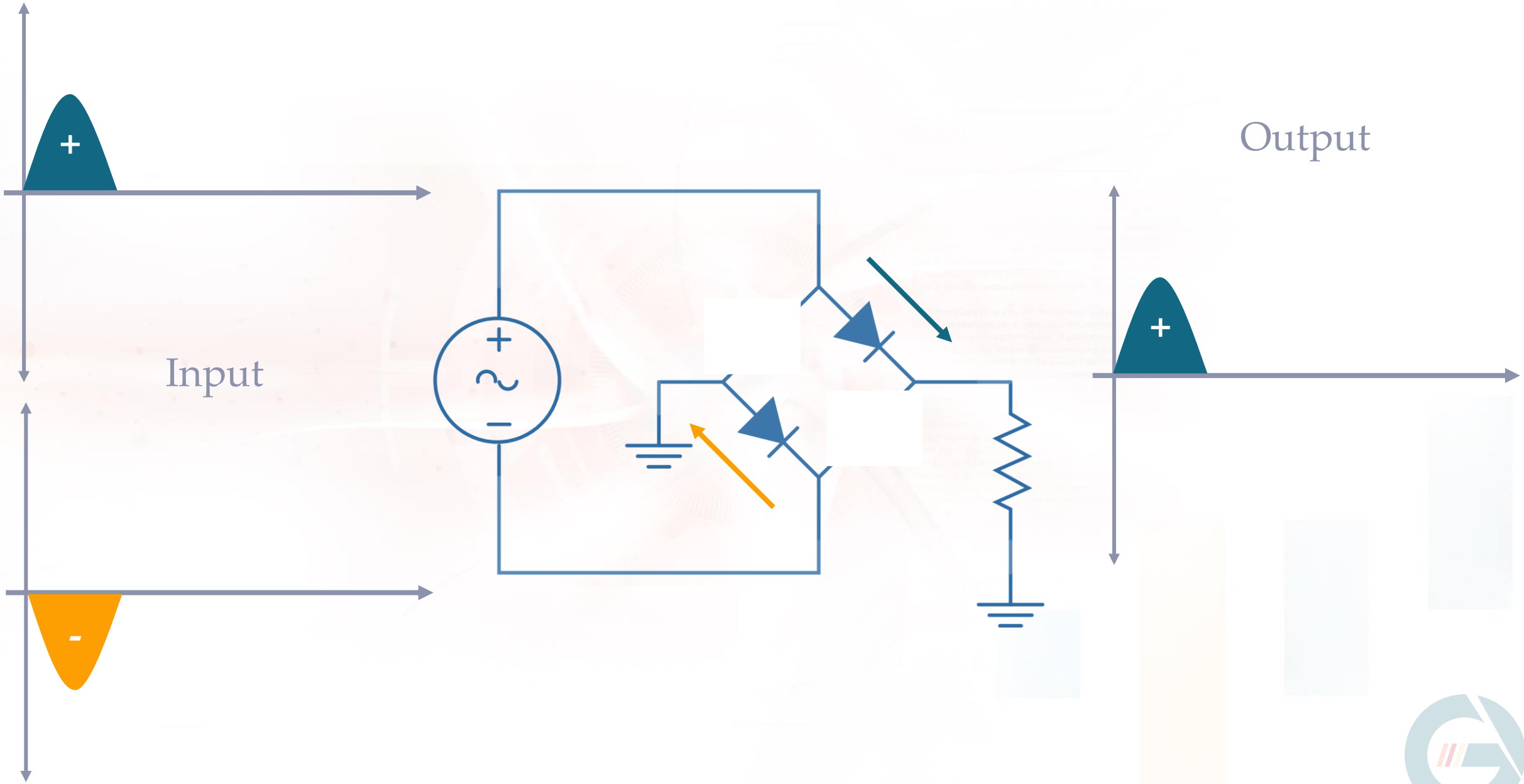


FULL-WAVE BRIDGE TYPE RECTIFIER



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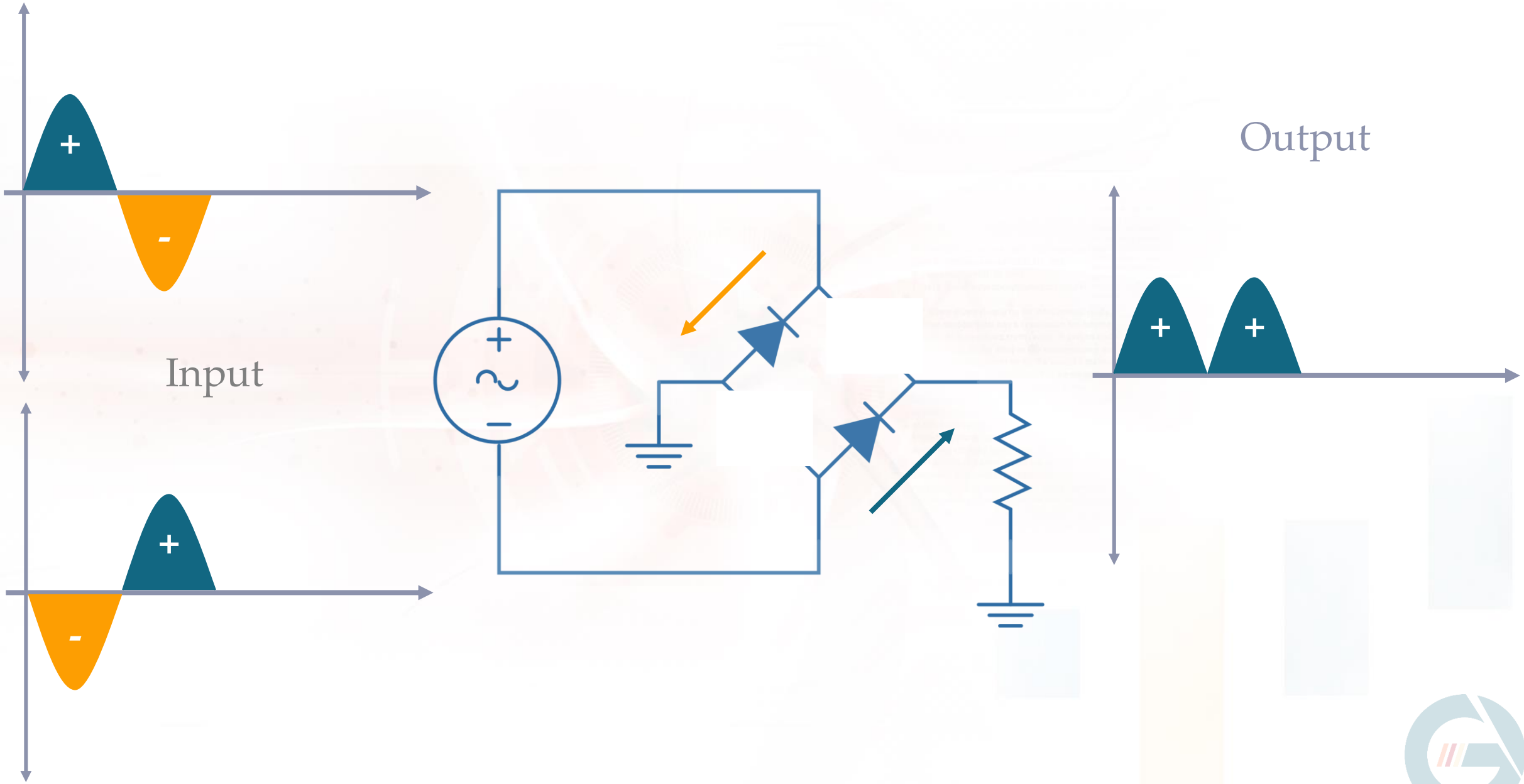

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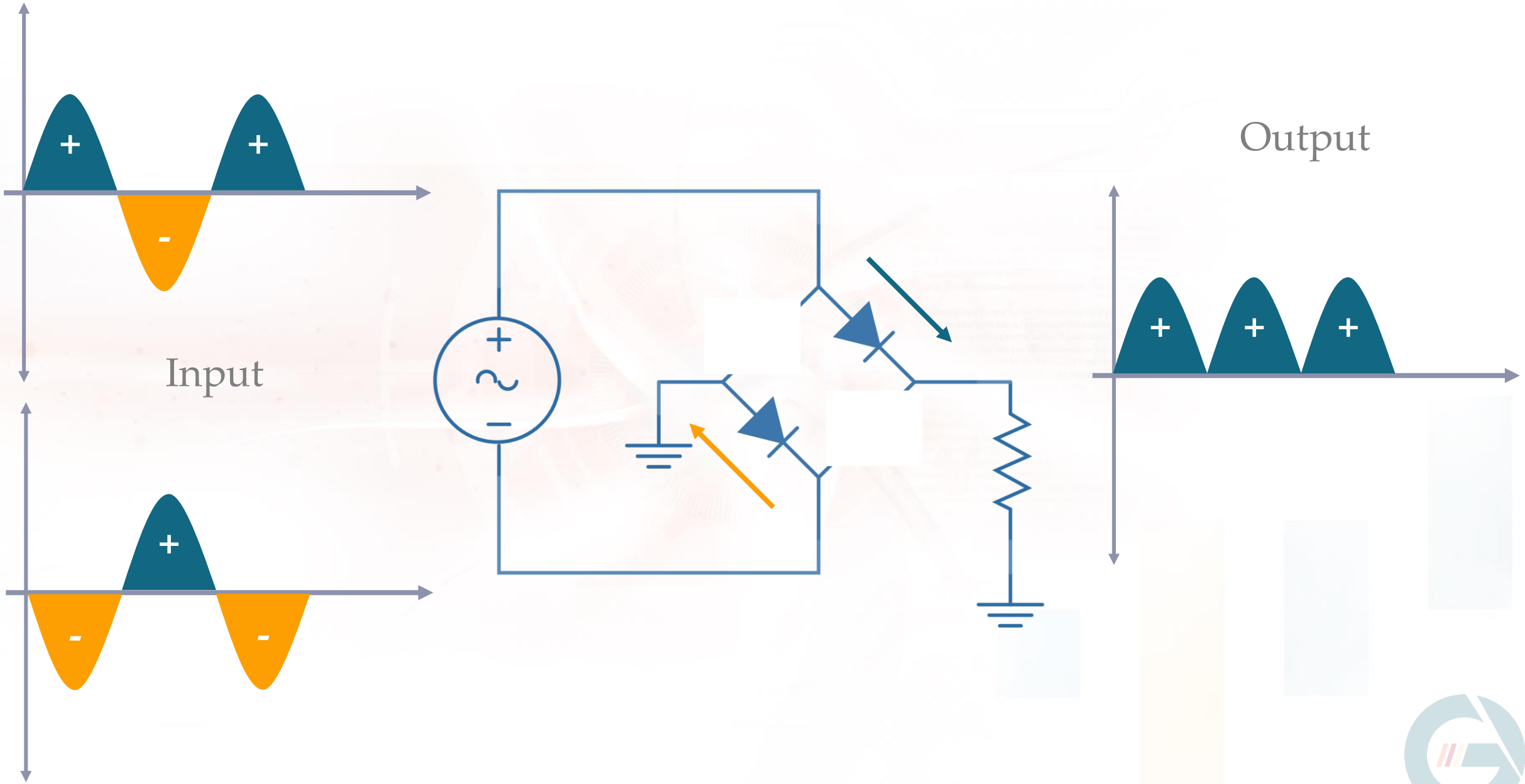

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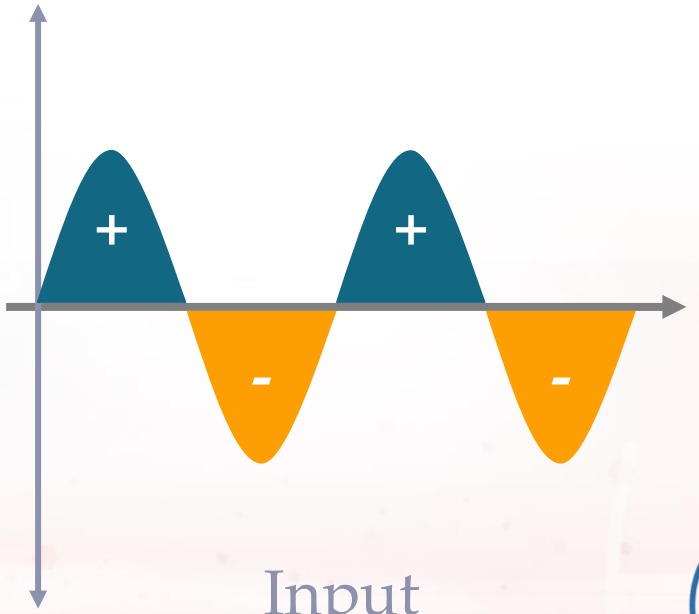

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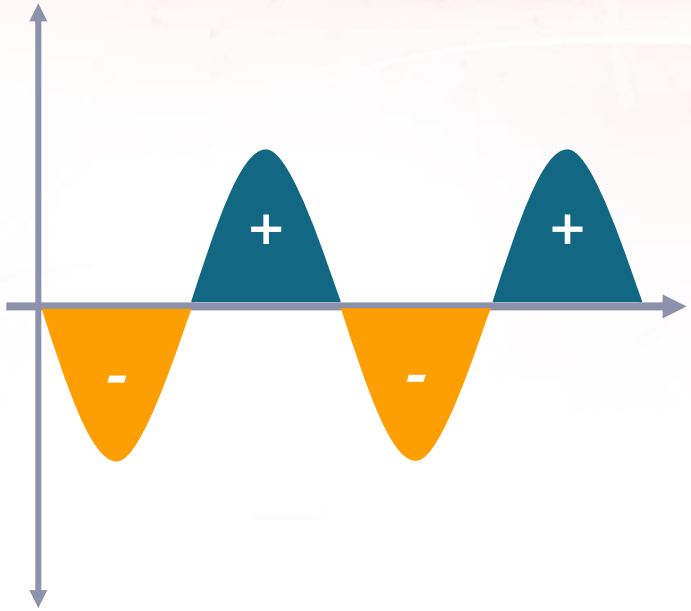

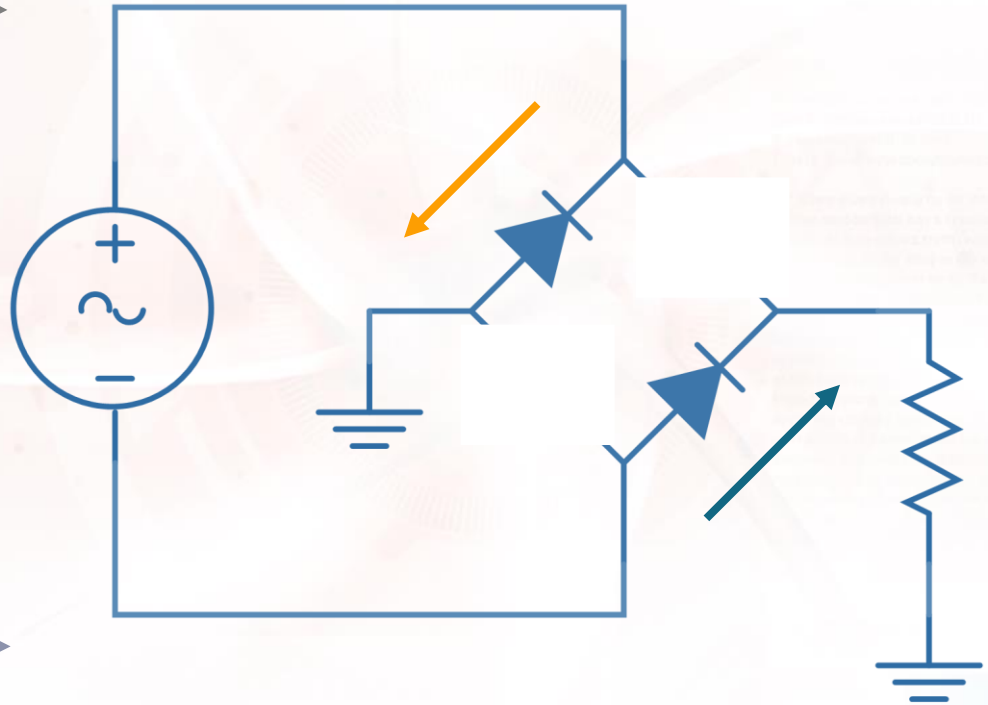

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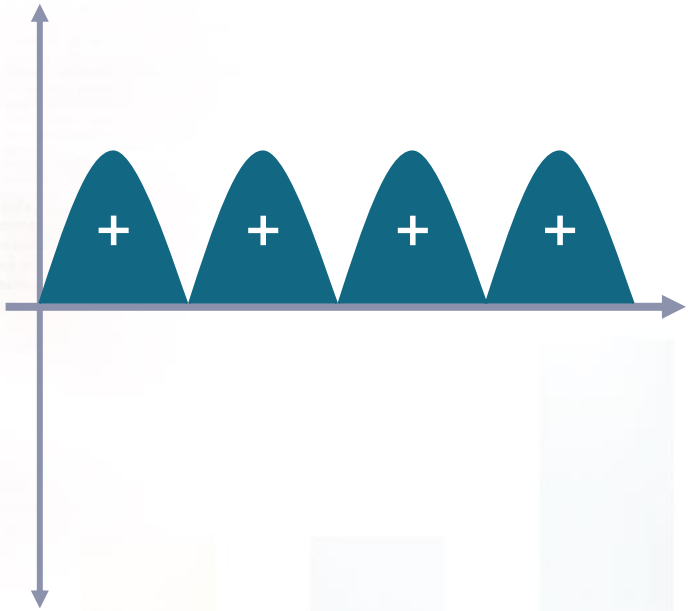

LABORATORY